

Moderna's 90-Day AI Transformation

Generative AI for Business Leaders & C-Suite | AI Implementation Case Study

1. Why Moderna's Story Matters

Moderna's AI adoption is not just another corporate technology success story. It is the clearest available proof that the 4-Phase AI Adoption Framework works at scale, in a heavily regulated industry, with measurable financial returns. Moderna went from zero AI usage to 3,000 employees using AI daily in 90 days, achieved ROI-positive status in four months, and generated a 300% return on investment in year one.

What makes Moderna's case particularly instructive is that the company operates in one of the most regulated industries on earth. Pharmaceutical and biotech companies face FDA oversight, clinical trial regulations, intellectual property constraints, and patient safety requirements that make many organisations dismiss rapid AI adoption as 'not possible in our industry.' Moderna proved that industry complexity is not a barrier to speed. It is a reason to move faster, because the competitive advantages of AI compound more rapidly in complex environments where information synthesis, document processing, and research acceleration are critical bottlenecks.

If Moderna can do this in biotech with 3,000 employees, your organisation can do it in your industry.

2. The Bold Start -- Universal Access from Day One

Moderna's first decision was its most consequential and most counterintuitive. Rather than restricting AI access to a small pilot group, the CEO announced AI as a strategic priority in an all-hands meeting and immediately provided ChatGPT Enterprise accounts to all 3,000 employees. Not just the data scientists. Not just the IT department. Every single person -- scientists, administrators, finance staff, HR, legal, regulatory affairs, marketing, facilities management, and executive assistants.

Key Point: Most organisations restrict AI access to a small group to 'manage risk.' Moderna did the opposite: universal access from Day 1. Their logic was simple -- the cost of 3,000 ChatGPT Enterprise licences (approximately \$2 million annually) was trivial compared to the innovation potential of 3,000 minds experimenting simultaneously. The real risk was not giving people access; it was being too slow.

This decision reflected a fundamental strategic insight: AI value is discovered, not designed. No executive team, no matter how visionary, can predict all the ways AI will improve operations across an organisation of 3,000 people doing hundreds of different jobs. The people closest to the work are the ones who know where the friction is. Give them the tools and permission, and they will find applications that no top-down initiative would have identified.

The psychology of universal access

Broad access also solved a change management problem. When only 50 people receive AI tools, the other 2,950 feel excluded. Exclusion breeds suspicion ('why do they get it and not us?'), resentment ('management is preparing to replace us'), and resistance ('I heard it does not work anyway'). When

everyone receives access simultaneously, AI becomes normalised. It is not a special programme for elite teams. It is a tool everyone gets, like email or a laptop.

3. The Discovery Phase -- 750 Use Cases in Four Weeks

Moderna's Weeks 2-4 were devoted to intensive discovery sessions across every department. These were not brainstorming workshops where ideas were written on sticky notes and promptly forgotten. They were hands-on working sessions where teams used AI on real tasks and documented what worked.

Department	Discovery Activities	Use Cases Identified
Research & Development	Scientists tested AI for literature reviews, research summaries, hypothesis generation, and experimental design documentation	180+
Regulatory Affairs	Teams tested AI for document analysis, compliance writing, submission preparation, and regulatory change tracking	120+
Marketing & Communications	Staff tested content creation, campaign development, internal communications, and market analysis	95+
Finance & Accounting	Teams tested report generation, variance analysis, budget modelling, and financial narrative writing	85+
Human Resources	Staff tested job description writing, candidate communications, onboarding materials, and policy documentation	75+
Legal	Teams tested contract review, risk assessment, legal research, and policy drafting	70+
IT & Operations	Staff tested technical documentation, troubleshooting guides, and process documentation	65+
Other departments	Various internal operations, facilities, procurement, executive support	60+

The total: 750 potential use cases identified organically by the people who do the work. This bottom-up discovery approach produced far more diverse and practical applications than any consulting engagement or executive workshop could have generated. Many of the most valuable use cases came from unexpected places -- an administrative assistant who figured out how to automate meeting preparation, a facilities manager who used AI to draft vendor communications, a research coordinator who streamlined literature review workflows.

4. From 750 to 50 -- The Art of Ruthless Prioritisation

Seven hundred fifty use cases sounds overwhelming. Moderna handled it with disciplined categorisation and ruthless prioritisation. They did not attempt to evaluate every idea individually. Instead, they categorised all 750 by department, potential impact, and implementation difficulty, then applied the three-criteria filter to identify the top 50 for immediate pilot testing.

Why 50, not 5 or 500?

Fifty simultaneous pilots seems aggressive, but the number was deliberate. With 3,000 employees, Moderna could staff 50 small pilot teams without overstretching any single department. Running 50 pilots simultaneously followed a portfolio theory of innovation: not every pilot would succeed, but with 50 experiments running in parallel, the probability of finding several breakthrough applications was extremely high. This is the opposite of the common 'pick one big pilot' approach that creates binary risk - either total success or total failure.

Real-World Incident -- Moderna's Internal Slack Channels (2023): Moderna created dedicated Slack channels where all 50 pilot teams shared discoveries in real time. When one team found an effective prompt or workflow, other teams adopted it within hours. A regulatory affairs team discovered that structuring AI prompts with specific document templates produced dramatically better outputs. Within 48 hours, teams across the company were using the same technique. Failures were shared openly without blame, which was equally valuable -- a marketing team's failed attempt to use AI for visual brand guidelines saved three other teams from repeating the experiment. This transparent learning culture accelerated organisational AI capability faster than any formal training programme could have.

5. The Results -- A Financial Case Study

By Week 9, Moderna had enough data to make clear scaling decisions. Fifteen of the fifty pilots showed exceptional results exceeding projections. These fifteen were immediately scaled across all relevant teams. The remaining thirty-five were either refined for continued testing or shelved without guilt.

Metric	Before AI	After AI (6 Months)	Impact
Drug development research productivity	Baseline	+30%	Scientists spent less time searching literature and more time on actual research
Regulatory document preparation time	Baseline	-50% (2x faster)	Directly accelerated product development pipeline and time to market
Annual AI tool cost (3,000 licences)	--	\$2 million	ChatGPT Enterprise at ~\$667 per employee per year
Measured productivity gains and cost savings	--	\$8 million+ (Year 1)	Across all scaled use cases combined
Return on Investment	--	300% in Year 1	Benefits compound as teams become more skilled
Time to ROI-positive	--	4 months	Cost recovered faster than most enterprise software deployments

The 300% ROI figure is striking, but it understates the true value. It measures only direct productivity gains and cost savings. It does not capture the strategic value of accelerated drug development timelines, which in biotech can mean billions in revenue from earlier market entry. Nor does it capture the competitive moat created by having 3,000 AI-literate employees who can leverage new AI capabilities as they emerge, while competitors are still figuring out their pilot programmes.

6. The Four Critical Lessons

Moderna's transformation distils into four lessons that apply to organisations of any size, in any industry.

Lesson 1: Move fast and give broad access from Day 1

The cost of broad AI access is minimal. ChatGPT Enterprise costs approximately \$60 per user per month. For a 500-person company, that is \$30,000 per month -- less than a single mid-level consultant. The innovation potential of 500 minds experimenting simultaneously is worth orders of magnitude more. Restricting access 'to manage risk' actually increases risk by slowing learning and concentrating knowledge in a small group that may not represent the organisation's real needs.

Lesson 2: Let employees discover use cases organically

Moderna's 750 use cases were not generated by a consulting firm or a strategy team. They were discovered by the people doing the work. This bottom-up approach works because: employees know their own pain points better than anyone else, they understand the nuances of their workflows that outsiders miss, they are more committed to solutions they helped discover, and their buy-in accelerates adoption because they feel ownership rather than imposition.

Lesson 3: Run many small pilots simultaneously

Fifty pilots running in parallel is a portfolio strategy. It accepts that some will fail in exchange for dramatically increasing the probability of finding breakthrough applications. Compare this to the common approach of running one carefully selected large pilot: if that single pilot fails, the entire programme stalls. With fifty pilots, even if thirty fail, the twenty that succeed provide a powerful foundation for scaling.

Lesson 4: Scale ruthlessly based on data

Moderna did not continue investing in mediocre results out of politeness or sunk cost thinking. Pilots that met or exceeded their metrics were scaled immediately. Pilots that underperformed were stopped. This discipline separates successful AI transformations from the endless pilot programmes that consume resources without delivering value. The willingness to kill underperforming initiatives quickly is just as important as the willingness to scale winners aggressively.

7. Replicating Moderna's Approach at Different Scales

Moderna is a \$50 billion+ company with 3,000 employees. The question many leaders ask is: 'Can I do this with 50 employees? With 200? With 10,000?' The answer is yes, with proportional adjustments.

Company Size	AI Tool Budget	Recommended Pilots	Timeline
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			Adjustment
10-50 employees	\$600-\$3,000/month	3-5 pilots, each with 2-3 people	Same 90-day framework
50-500 employees	\$3,000-\$30,000/month	5-15 pilots, each with 3-5 people	Same 90-day framework
500-5,000 employees	\$30,000-\$300,000/month	15-30 pilots across departments	Same 90-day framework
5,000+ employees	\$300,000+/month	30-50+ pilots, department-level coordinators	90-day framework with regional phasing

The 90-day framework does not change with company size. What changes is the number of simultaneous pilots, the coordination overhead, and the total tool budget. But the core principle remains: universal access, bottom-up discovery, parallel pilots, data-driven scaling.

8. Common Mistakes When Copying Moderna's Model

Organisations often try to replicate Moderna's success but miss critical elements. These mistakes reduce the approach from transformational to mediocre.

- **Providing access without permission to experiment:** Giving everyone AI licences but not explicitly telling them to use the tools on real work creates an 'available but unused' situation. Moderna's CEO did not just provide tools; he publicly declared that experimentation was expected and that failure was acceptable.
- **Running discovery sessions as brainstorming rather than hands-on work:** Moderna's teams tested AI on actual tasks during discovery. Many organisations run workshops where people imagine what AI could do without actually trying it. Imagination produces optimistic but untested ideas. Hands-on testing produces practical, proven applications.
- **Treating pilot failures as programme failures:** When 5 of 50 pilots fail, some organisations panic and slow down. Moderna expected failures and used them as learning opportunities shared transparently. The psychological safety to fail fast and share openly is essential.
- **Scaling by committee rather than by data:** Moderna's scaling decisions were binary and data-driven. Some organisations form committees to 'evaluate' successful pilots before scaling, introducing months of delay. If a pilot met its pre-defined metrics, it scales. Period.

9. The Competitive Implications

Moderna's 90-day transformation created a compound competitive advantage that becomes harder to replicate over time. By the time a competitor begins their own AI transformation, Moderna's employees have six months of daily AI experience. They have refined their prompts, integrated AI into standard workflows, and developed institutional knowledge about what works. This experience gap widens every month.

In biotech specifically, speed-to-insight has direct revenue implications. If AI helps Moderna identify a promising drug candidate three months faster than competitors, that three-month advantage translates

to earlier patent filing, earlier clinical trials, earlier market entry, and potentially billions in first-mover revenue. The \$2 million annual investment in AI tools is trivial compared to the strategic value of accelerated discovery.

This dynamic applies equally in other industries. A financial services firm where every analyst uses AI is producing better research faster than a competitor where AI is locked in a pilot programme. A retail company where every buyer uses AI for market analysis is spotting trends earlier than competitors relying on manual research. Speed compounds. The gap accelerates.

10. Key Terms Glossary

Term	Definition
Universal access	The strategy of providing AI tools to every employee simultaneously, rather than restricting access to select teams or departments.
Bottom-up discovery	An approach where use cases are identified by frontline employees experimenting with AI on their actual work, rather than being prescribed by leadership or consultants.
Portfolio theory of innovation	Running many small experiments in parallel to maximise the probability of finding breakthrough applications, accepting that some experiments will fail.
Ruthless prioritisation	Making binary scale/stop decisions based on data, without continuing mediocre projects out of politeness, political considerations, or sunk cost thinking.
Compound competitive advantage	An advantage that grows wider over time as experience, institutional knowledge, and workflow integration accumulate, making it progressively harder for competitors to catch up.
Transparent learning culture	An organisational environment where both successes and failures are shared openly without blame, accelerating collective learning across teams.
ROI-positive status	The point at which cumulative benefits from AI adoption exceed cumulative costs, achieved by Moderna in just four months.
Experience gap	The widening difference in AI capability between early adopters (whose employees gain daily AI experience) and late movers (whose employees have not started).

11. Quick Revision Checklist

- Moderna gave all 3,000 employees AI access from Day 1, investing ~\$2 million annually in ChatGPT Enterprise licences.
- Bottom-up discovery produced 750 use cases in 4 weeks -- far more than any top-down initiative could have identified.
- 50 pilots launched simultaneously in Week 5, each with a clear owner, defined metrics, and 4-week timelines.
- 15 of 50 pilots showed exceptional results and were scaled immediately; 35 were refined or shelved.

- Drug development productivity increased 30%; regulatory document preparation became 50% faster.
- ROI-positive in 4 months; 300% return on investment in Year 1 (\$8M+ value against \$2M cost).
- Four critical lessons: broad access, organic discovery, parallel pilots, data-driven scaling.
- Universal access solves change management by normalising AI -- exclusion breeds suspicion and resistance.
- Internal Slack channels for transparent sharing accelerated learning faster than formal training.
- The framework scales from 10 employees to 10,000+ with proportional adjustments to pilot count and budget.
- Speed creates compound competitive advantage -- the experience gap widens every month.

20 Exam-Based MCQs — Moderna's AI Transformation

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. How many employees did Moderna give AI access to on Day 1?

- A) 50 senior executives
- B) 500 researchers and scientists only
- C) All 3,000 employees across every department
- D) 1,000 employees selected by department heads

Answer: C **Explanation:** C is correct -- Moderna provided ChatGPT Enterprise access to all 3,000 employees from Day 1, including scientists, administrators, finance, HR, legal, and every other department. A, B, and D all describe restricted access approaches that Moderna deliberately rejected in favour of universal access.

Q2. *Scenario:* A manufacturing company's CEO wants to replicate Moderna's approach. The CFO argues that providing AI tools to all 800 employees at \$60 per user per month (\$576,000 annually) is too expensive and proposes limiting access to a 30-person innovation team instead. What is the STRONGEST argument against the CFO's proposal, based on Moderna's experience?

- A) The innovation team might select the wrong use cases
- B) Moderna's experience shows that bottom-up discovery by all employees produces far more diverse and practical use cases than top-down selection by a small team, and the \$576K investment is trivial if even a fraction of 800 employees find efficiency gains
- C) Other employees will feel excluded and resist AI later
- D) The innovation team will become an organisational bottleneck

Answer: B **Explanation:** B is correct because it addresses both the strategic value (bottom-up discovery from 800 minds versus 30) and the financial argument (Moderna's \$2M investment returned \$8M+ in Year 1, suggesting that even a fraction of similar gains from 800 employees would exceed \$576K). C and D identify real but secondary concerns. A is partially relevant but less compelling than the combined strategic and financial argument.

Q3. How many potential AI use cases did Moderna employees identify during the discovery phase?

- A) 50
- B) 150
- C) 500
- D) 750

Answer: D **Explanation:** D is correct -- Moderna employees identified 750 potential use cases across all departments within four weeks. This large number resulted from universal access and bottom-up discovery rather than top-down identification.

Q4. *Scenario:* A professional services firm has given all 200 employees AI access and run discovery sessions. They identified 120 potential use cases. Leadership wants to select 5 pilots for careful, sequential testing -- one at a time, each running for 8 weeks before deciding on the next. What does Moderna's experience suggest about this approach?

- A) It is the safest approach and should be followed
- B) Sequential testing is unnecessarily slow -- running 10-15 pilots simultaneously (proportional to company size) increases the probability of finding breakthrough applications and compresses the learning timeline
- C) 5 pilots is too few; all 120 should be tested
- D) The firm should wait until they have 750 use cases before starting

Answer: B **Explanation:** B is correct -- Moderna ran 50 pilots simultaneously with 3,000 employees. Proportionally, a 200-person firm could comfortably staff 10-15 parallel pilots. Sequential testing wastes time because each pilot waits for the previous one to complete, and failures in early pilots can discourage continuing. A is overly cautious. C is impractical. D applies Moderna's specific number without scaling appropriately.

Q5. What was Moderna's approximate return on investment from AI adoption in Year 1?

- A) 50%
- B) 150%
- C) 300%
- D) 500%

Answer: C **Explanation:** C is correct -- Moderna invested approximately \$2 million annually in AI tools and measured \$8 million+ in productivity gains and cost savings in Year 1, representing a 300% ROI.

Q6. *Scenario:* During Moderna's pilot phase, a regulatory affairs team discovered that structuring AI prompts with specific document templates produced dramatically better outputs. They shared this technique on an internal Slack channel. Within 48 hours, teams across multiple departments were using the same technique. What principle does this example BEST illustrate?

- A) The importance of formal AI training programmes
- B) The value of transparent learning culture where discoveries are shared openly and adopted rapidly across the organisation
- C) The need for standardised AI prompts mandated by IT
- D) The superiority of regulatory teams in AI adoption

Answer: B **Explanation:** B is correct -- this example illustrates how transparent sharing of discoveries (both successes and failures) accelerates organisational learning faster than formal training. The Slack channel enabled peer-to-peer knowledge transfer at speed. A describes formal programmes, which Moderna supplemented with organic sharing. C implies top-down mandates, which contradicts the organic nature of the discovery. D overgeneralises from one team's contribution.

Q7. Of Moderna's 50 simultaneous pilots, how many showed exceptional results and were immediately scaled?

- A) 5
- B) 15
- C) 35
- D) 50

Answer: B **Explanation:** B is correct -- 15 of the 50 pilots showed exceptional results exceeding projections and were immediately scaled across all relevant teams. The remaining 35 were either refined for continued testing or shelved. This 30% success rate demonstrates why running many parallel pilots (portfolio approach) is superior to betting on a single pilot.

Q8. *Scenario:* A biotech company is replicating Moderna's approach. After 6 weeks, 8 of their 20 pilots are showing strong results, 5 are showing marginal improvement, and 7 have produced no measurable impact. The Head of Innovation wants to continue all 20 pilots for another 6 weeks 'to give every team a fair chance.' What would Moderna's approach recommend?

- A) Continue all 20 pilots as the Head of Innovation suggests
- B) Scale the 8 successful pilots immediately, give the 5 marginal pilots one more short iteration with adjusted parameters, and shelve the 7 that produced no results
- C) Stop all pilots and redesign the programme from scratch
- D) Focus only on the single most successful pilot and abandon the rest

Answer: B **Explanation:** B is correct -- Moderna's approach calls for ruthless, data-driven scaling decisions. Successful pilots scale immediately, marginal pilots get one more short iteration (not six more weeks), and non-performers are shelved without guilt. A perpetuates mediocre results. C wastes proven successes. D is too conservative and wastes seven other successful pilots.

Q9. What does 'compound competitive advantage' mean in the context of AI adoption?

- A) The financial returns from AI investments grow at a compound interest rate

- B) The experience gap between early and late AI adopters widens over time because daily AI usage builds institutional knowledge and workflow integration that competitors starting later cannot quickly replicate
- C) Companies that adopt AI first can charge compound premiums to customers
- D) AI tools become exponentially cheaper over time

Answer: B **Explanation:** B is correct -- compound competitive advantage means the gap between early adopters and late movers widens every month. While an early adopter's employees refine prompts, integrate AI into workflows, and develop institutional AI knowledge through daily use, a competitor just starting their pilot programme is months or years behind and falling further behind with each passing week. A, C, and D describe different concepts.

Q10. Scenario: A retail company with 150 employees wants to replicate Moderna's model. Their annual AI tool budget is \$108,000 (\$60/user/month x 150 employees x 12 months). The CEO asks: 'How do we know this investment will pay off?' Based on Moderna's ROI data, what is the MOST compelling response?

- A) There are no guarantees, but AI is important for the future
- B) Moderna achieved 300% ROI in Year 1 (\$8M return on \$2M investment). Even if your company achieves one-tenth of Moderna's efficiency per employee, the \$108K investment would return \$400K+ in productivity gains -- nearly 4x the cost
- C) The investment will pay off within one week
- D) AI tools are a tax-deductible expense, so the effective cost is lower

Answer: B **Explanation:** B is correct because it uses Moderna's verified data to build a conservative financial case. Moderna's per-employee return was approximately \$2,667 (\$8M / 3,000 employees). Even at one-tenth efficiency (\$267 per employee), 150 employees would generate approximately \$40K in savings -- and the actual number is likely much higher than one-tenth. A is vague and unconvincing. C is unrealistic. D is technically true but financially irrelevant to the ROI question.

Q11. What was the primary change management benefit of Moderna's universal access approach?

- A) It reduced the need for training programmes
- B) It normalised AI as a standard tool for everyone, avoiding the suspicion and resistance that arise when access is restricted to a select group
- C) It ensured that only interested employees used the tools
- D) It allowed the IT department to control usage patterns

Answer: B **Explanation:** B is correct -- universal access normalised AI by making it a standard tool like email or a laptop. Restricted access breeds suspicion ('why them and not us?'), resentment, and resistance. When everyone receives access simultaneously, AI adoption becomes a shared organisational experience rather than a divisive special programme. A is incorrect -- training was still provided. C describes a passive approach, not what Moderna did (they actively expected experimentation). D contradicts the decentralised nature of Moderna's approach.

Q12. Scenario: A financial services company has completed its 90-day AI transformation. A new competitor enters the market and begins their own AI adoption programme. The CEO asks whether the competitor can catch up quickly. Based on the concept of compound competitive advantage, what is the MOST accurate assessment?

- A) The competitor can catch up within 30 days by following the same framework
- B) The competitor will face a widening gap because the early adopter's employees have months of daily AI experience, refined workflows, and institutional knowledge that cannot be quickly replicated
- C) Technology levels the playing field, so the competitor will reach parity once they adopt the same tools
- D) The competitive advantage only applies in technology industries

Answer: B **Explanation:** B is correct -- compound competitive advantage means the gap widens over time. While the competitor starts discovery sessions, the early adopter's employees are already optimising their second and third generation of AI workflows. The institutional knowledge, refined prompts, and embedded processes create an advantage that grows with each month of experience. A underestimates the knowledge gap. C ignores the human expertise dimension. D incorrectly limits the concept to one industry.

Q13. How long did it take Moderna to achieve ROI-positive status on their AI investment?

- A) 2 months
- B) 4 months
- C) 8 months
- D) 12 months

Answer: B **Explanation:** B is correct -- Moderna achieved ROI-positive status in just four months, meaning the cumulative productivity gains and cost savings exceeded the cumulative cost of AI tools by the four-month mark. This was faster than most traditional enterprise software deployments achieve payback.

Q14. Scenario: A company's legal department refuses to participate in the AI discovery phase, arguing that legal work is 'too sensitive' for AI. Meanwhile, Moderna's legal team used AI for contract review, risk assessment, legal research, and policy drafting during their transformation. What is the BEST approach to address the legal department's resistance?

- A) Exempt the legal department entirely and focus on other departments
- B) Share Moderna's legal team example to demonstrate that legal work can benefit from AI, start with low-risk internal legal tasks (policy drafts, research memos) rather than client-facing work, and address specific data privacy concerns directly
- C) Override the legal department and mandate participation
- D) Wait until other departments have proven AI's value before approaching legal again

Answer: B **Explanation:** B is correct because it addresses the legal team's concerns with evidence (Moderna's legal team used AI successfully), proposes a low-risk starting point that respects sensitivity concerns, and directly addresses data privacy. A leaves a significant department behind and sets a

precedent for opt-outs. C creates resentment and poor adoption. D delays unnecessarily and allows resistance to harden.

Q15. What was the approximate annual cost of providing ChatGPT Enterprise to all 3,000 Moderna employees?

- A) \$500,000
- B) \$2 million
- C) \$10 million
- D) \$50 million

Answer: B **Explanation:** B is correct -- the lecture states that the cost of ChatGPT Enterprise for 3,000 employees was approximately \$2 million annually (roughly \$667 per employee per year or ~\$56 per employee per month).

Q16. *Scenario:* A technology company is in Week 5 of their AI transformation. Of 20 running pilots, 3 have already failed to produce measurable results. The CEO is concerned and wants to pause the programme to 'understand what went wrong before continuing.' How would Moderna's approach handle this situation?

- A) Pause the programme as the CEO suggests
- B) Continue the remaining 17 pilots, shelve the 3 failures, analyse what went wrong for organisational learning without stopping progress, and share the failure analysis transparently
- C) Replace the 3 failed pilots with 3 new ones immediately
- D) Reduce the total number of pilots to prevent further failures

Answer: B **Explanation:** B is correct -- Moderna expected some pilots to fail and treated failures as learning opportunities shared transparently without blame. Stopping the programme because 3 of 20 pilots failed (a 15% failure rate) would sacrifice the 17 that may succeed. A overreacts to expected failures. C adds complexity without learning from the failures. D reduces the portfolio size and probability of finding breakthroughs.

Q17. What area saw a 50% improvement in speed at Moderna due to AI adoption?

- A) Employee onboarding
- B) Regulatory document preparation
- C) Sales cycle length
- D) Manufacturing processes

Answer: B **Explanation:** B is correct -- the lecture states that regulatory document preparation became 50% faster (2x speed improvement) after AI adoption, which directly accelerated Moderna's product development pipeline and time to market. A, C, and D were not cited as specific 50% improvement areas.

Q18. Scenario: A healthcare company with 400 employees is planning its AI adoption. Using the scaling table from the notes, they plan to run 5-15 pilots with 3-5 people each. However, their HR director insists on developing a comprehensive 6-month training programme before any employee touches an AI tool. What does Moderna's experience suggest about this training-first approach?

- A) Comprehensive training is essential and should not be skipped
- B) Moderna's approach prioritised hands-on experimentation with real work over formal training, finding that employees learn AI most effectively by using it on actual tasks rather than through theoretical coursework
- C) Training is unnecessary if employees have access to AI tools
- D) The HR director should design the training but compress it to 3 months

Answer: B **Explanation:** B is correct -- Moderna provided basic training but emphasised hands-on discovery with real work tasks. Their experience showed that employees learned most effectively through experimentation, with knowledge shared organically through peer channels. A delays action and prioritises theory over practice. C goes too far in dismissing training entirely. D still creates unnecessary delay.

Q19. What does the 'portfolio theory of innovation' mean in the context of AI pilots?

- A) Investing in AI stocks to diversify a financial portfolio
- B) Running many small experiments in parallel to maximise the probability of finding breakthrough applications, accepting that some will fail
- C) Creating a portfolio of AI tools from different vendors
- D) Documenting all AI use cases in a portfolio for board presentations

Answer: B **Explanation:** B is correct -- the portfolio approach to innovation means running many parallel experiments (Moderna ran 50) to increase the statistical probability of finding exceptional applications. Some will fail, but the winners more than compensate. This is the opposite of betting everything on a single large pilot. A confuses innovation with financial investing. C and D use the word 'portfolio' in unrelated contexts.

Q20. Scenario: After completing a successful 90-day AI transformation, a company's board asks the CEO to quantify the 'strategic value' of AI beyond the measurable productivity gains. The CEO knows that productivity gains are only part of the story. Which of the following represents strategic value that Moderna's case demonstrates but that is NOT captured in the 300% ROI calculation?

- A) The cost of AI tool licences
- B) Accelerated drug development timelines that could translate to earlier market entry and potentially billions in first-mover revenue, plus 3,000 AI-literate employees who can leverage new capabilities as they emerge
- C) Employee satisfaction improvements from reduced administrative work
- D) Reduced IT support tickets related to AI tool issues

Answer: B **Explanation:** B is correct -- the lecture explicitly states that the 300% ROI figure 'understates the true value' because it does not capture the strategic value of accelerated drug development timelines (which in biotech can mean billions) or the competitive moat created by 3,000 AI-literate employees. A is a

cost, not a value. C is a real but secondary benefit. D is operationally trivial.